

Cross-Dataset Intrusion Detection

Bridging the Generalization Gap in ML-Based NIDS through Bidirectional Evaluation & Domain Alignment

Department: Computer Science & Engineering	Institution: USTM, Meghalaya
Guide: Mr. Abdul Kayum Ali (Asst. Prof.)	Supervisor: Mr. Debarbat Bharali (Asst. Prof.)
Deck: final.pptx (12 Slides)	Session: 2026–2027

1. Team Slide Allocation (final.pptx — 12 Slides)

The presentation is divided equally into 4 equal segments of **3 slides each** (~3.5 minutes per member), matching each member's engineering responsibility:

Member & Roll No.	Slides	Core Responsibility & Slide Topics	Time
1. Pynskhemlang Marbaniang 2023/BCSE/0005	Slides 1–3	Title, Background (NIDS, ML/DL), Problem Statement (Same-Dataset Evaluation Illusion)	3.5 min
2. Tehjib Almas Junaid 2023/BCSE/0032	Slides 4–6	Importance, Scope of Study (2 datasets, 5 models), Literature Review (Verkerken, Cantone, Ahmim, Bertoli, Sarhan)	3.5 min
3. Bamelaa Marbaniang 2023/BCSE/0045	Slides 7–9	3 Research Gaps (Diagnosis vs Remediation), 4-Pillar Feasibility (0 budget), 4 Formal Objectives	3.5 min
4. Md Sabique Huda 2023/BCSE/0066	Slides 10–12	7-Step Methodology Pipeline, 9-Week Timeline, 5 Expected Deliverables, References & Closing Q&A;	3.5 min

2. Slide-by-Slide Presentation Script

SLIDE 1: Title Slide

Speaker 1: Pynskhemlang Marbaniang

Visual & Focus: Introduce title, 4 team members, guide Mr. Abdul Kayum Ali, supervisor Mr. Debarbat Bharali, and USTM.

Spoken Script: "Respected Chairperson, honourable guide Mr. Abdul Kayum Ali sir, project supervisor Mr. Debarbat Bharali sir, and faculty members. A very good morning to all. Today we present our Minor Project Synopsis: 'Cross-Dataset Intrusion Detection: Bridging the Generalization Gap in ML-Based NIDS through Bidirectional Cross-Dataset Evaluation and Domain Alignment'. Our team members are Pynskhemlang Marbaniang, Tehjib Almas Junaid, Bamelaai Marbaniang, and Md Sabique Huda from the Department of CSE, USTM. Our research addresses why ML intrusion detection models that achieve 99% accuracy in research collapse down to 40–60% when deployed into real-world networks."

Transition Cue: [Action: Click to Slide 2]

SLIDE 2: 1. Introduction — i. Background of the Study

Speaker 1: Pynskhemlang Marbaniang

Visual & Focus: Network Intrusion Detection Systems (NIDS), signature vs ML/DL, 98%–99.5% accuracy claims.

Spoken Script: "Network Intrusion Detection Systems (NIDS) are the frontline defense inspecting network traffic in real time for exploits, protocol violations, and data exfiltration. Historically, signature systems like Snort, Suricata, and Zeek matched traffic against known rules — precise on known exploits, but blind to zero-day attacks and novel threats. Therefore, cybersecurity researchers shifted to Machine Learning and Deep Learning (Decision Trees, Random Forests, SVM, XGBoost). In published academic papers, these models routinely report 98% to 99.5% accuracy on benchmarks such as CICIDS2017 and UNSW-NB15."

Transition Cue: [Action: Click to Slide 3]

SLIDE 3: ii. Problem Statement — The Same-Dataset Evaluation Illusion

Speaker 1: Pynskhemlang Marbaniang

Visual & Focus: In the Lab (closed world, memorizes artifacts) vs In Deployment (covariate shift, accuracy collapses 99% -> near-random).

Spoken Script: "However, this near-perfect accuracy hides a systemic failure: 'The Same-Dataset Evaluation Illusion'. In the lab, models are trained and tested on splits from the exact same dataset. They share identical network topologies, background noise, tooling, and time windows — a closed world. The model memorizes simulator artifacts rather than real attack signatures. When deployed into a different network — the cross-dataset setting — the model faces severe covariate shift and concept drift. Accuracy plummets from 99% down toward random guessing (40%–60%), and Matthews Correlation Coefficient collapses to near-zero. I now hand over to Tehjib Almas Junaid to discuss the scope and literature review."

Transition Cue: [Handover to Tehjib • Action: Click to Slide 4]

SLIDE 4: iii. Importance of the Present Work & iv. Scope of the Study

Speaker 2: Tehjib Almas Junaid

Visual & Focus: Why it matters (deployed false confidence); 2 Datasets, 5 Classifiers, Bidirectional Transfer, Mitigation Layer.

Spoken Script: "Thank you, Pynskhemlang. Deployed NIDS that fail under real-world drift create false confidence while leaving networks exposed. Closing this gap is essential for trustworthy, deployable ML security. Our project scope spans four pillars: First, 2 benchmark datasets: CICIDS2017 (80+ features) and UNSW-NB15 (49 features). Second, 5 classifier families: Decision Tree, Random Forest, Logistic Regression, SVM, and XGBoost. Third, bidirectional transfer: testing both CICIDS → UNSW and UNSW → CICIDS. Fourth, an active mitigation layer: engineering feature selection, robust normalization, and CORAL alignment."

Transition Cue: [Action: Click to Slide 5]

SLIDE 5: 2. Literature Review (Part 1)

Speaker 2: Tehjib Almas Junaid

Visual & Focus: Table of prior literature: Verkerken et al. (2022), Cantone et al. (2024), Ahmim et al. (2024).

Spoken Script: "We synthesized cornerstone literature in cross-dataset NIDS. First, Verkerken et al. (2022) in the Journal of Network & Systems Management evaluated bidirectional transfer between CICIDS2017 and UNSW-NB15, proving that accuracy plummets 28% to 42% and stating that feature harmonization is mandatory. Second, Cantone et al. (2024) at IEEE CSR tested pairwise transfer across 4 CIC-family datasets, showing that cross-dataset MCC collapses from over 99.6% to near-zero. Third, Ahmim et al. (2024) evaluated UNSW-NB15 and TON_IoT, finding that a naive intersection of only 7 features caused accuracy to decline by up to 35%."

Transition Cue: [Action: Click to Slide 6]

SLIDE 6: 2. Literature Review (Part 2)

Speaker 2: Tehjib Almas Junaid

Visual & Focus: Bertoli et al. (2022) and Sarhan et al. (2021).

Spoken Script: "Continuing the review: Fourth, Bertoli et al. (2022) evaluated DoS models transferred from CICIDS2017 onto an independent testbed, logging a 25.63% accuracy drop because models rely on non-transferable background features. Fifth, Sarhan et al. (2021) examined standardized NetFlow extraction across 4 benchmarks, confirming that feature-set heterogeneity is the primary obstacle to generalization. I now invite Bamelaai Marbianang to present the research gaps, feasibility, and formal objectives."

Transition Cue: [Handover to Bamelaai • Action: Click to Slide 7]

SLIDE 7: 3. Research Gap — Three Pillars

Speaker 3: Bamelaai Marbaniang

Visual & Focus: Pillar I (Diagnosis vs Remediation), Pillar II (Feature Ontology Mismatch), Pillar III (No Bidirectional Benchmark).

Spoken Script: "Thank you, Tehjib. The reviewed literature exposes three critical research gaps: Pillar I: 'Diagnosis vs. Remediation'. Existing studies — including Verkerken et al. and Cantone et al. — observed the collapse and treated it as an observational dead-end. Our project moves from observational diagnosis to active engineering remediation: we actively engineer solutions to bridge the gap! Pillar II: 'Semantic Feature Ontology Mismatch'. CICIDS (CICFlowMeter, 80+ features) and UNSW (Argus/Zeek, 49 features) use incompatible naming, scales, and definitions with no existing alignment ontology. Pillar III: 'No Bidirectional Robustness Benchmark'. Prior works test only one direction and never compare how tree-based vs margin-based classifiers withstand domain shift."

Transition Cue: [Action: Click to Slide 8]

SLIDE 8: 4. Feasibility Study

Speaker 3: Bamelaai Marbaniang

Visual & Focus: Technical (open data, <6 GB RAM), Economic (■0.00), Operational (4 modules), Schedule (8–10 weeks).

Spoken Script: "We evaluated four feasibility dimensions: Technical: Both datasets are open-access from university portals; models are supported in Scikit-Learn and XGBoost; streaming and stratified subsampling (200k–500k records) constrain RAM under 6 GB. Economic: The entire software toolchain is open-source, and experiments execute on student laptops and free Google Colab GPUs. Net expenditure is ■0.00. Operational: The project maps directly onto 4 modular tasks for our 4 team members via GitHub. Schedule & Risk: Planned over 8–10 weeks; class imbalance is resolved through stratified sampling and SMOTE-NC; feature scales via Robust and Quantile normalization."

Transition Cue: [Action: Click to Slide 9]

SLIDE 9: 5. Objectives

Speaker 3: Bamelaai Marbaniang

Visual & Focus: 4 formal objectives (Analyze failure, Design testbed, Benchmark 5 classifiers, Mitigation & Dashboard).

Spoken Script: "Our project pursues four clear objectives: 1. Analyze and mathematically formulate why single-dataset cross-validation fails (covariate shift, concept drift, shortcut artifacts). 2. Design a bidirectional testbed harmonizing CICIDS2017 and UNSW-NB15 into a shared, semantically aligned feature space. 3. Benchmark 5 classifiers on same-dataset and cross-dataset regimes using Accuracy, Precision, Recall, Macro-F1, MCC, and FAR. 4. Apply feature selection and CORAL domain-alignment to reduce the generalization gap, and deploy an interactive Streamlit dashboard. I now invite Md Sabique Huda to present our methodology, deliverables, and timeline."

Transition Cue: [Handover to Md Sabique • Action: Click to Slide 10]

SLIDE 10: 6. Methodology / Planning of Work

Speaker 4: Md Sabique Huda

Visual & Focus: 7-step pipeline (Acquisition -> Mapping -> Baselines -> Transfer -> Alignment -> Retrain -> Dashboard) + 9-week timeline.

Spoken Script: "Thank you, Bamelaai. Our methodology follows an end-to-end 7-step pipeline: Step 1: Clean raw CSVs, handle division-by-zero infinite flow rates, and impute missing data. Step 2: Build the semantic mapping dictionary linking equivalent features (e.g., CICIDS 'Flow Duration' ↔ UNSW 'dur'). Step 3: Establish intra-dataset baseline ceilings (97%–99%). Step 4: Execute zero-shot bidirectional cross-dataset transfer (CICIDS ↔ UNSW), exposing the drop. Step 5: Apply Mutual Information feature selection and CORAL covariance alignment without target labels. Step 6: Retrain models on aligned features and calculate the Generalization Recovery Ratio (GRR). Step 7: Deploy an interactive Streamlit dashboard. As shown in our timeline: Weeks 1–4 are led by Pynskhemlang and Tehjib, Weeks 4–7 by Tehjib, Sabique, and Bamelaai, and Weeks 7–9 by all members."

Transition Cue: [Action: Click to Slide 11]

SLIDE 11: 7. Expected Outcome

Speaker 4: Md Sabique Huda

Visual & Focus: 5 Deliverables: Evaluation Harness, Generalization Map, Alignment Toolkit, Robustness Ranking, Streamlit Dashboard.

Spoken Script: "Our project delivers five concrete outcomes: 1. A Bidirectional Evaluation Harness: Automated Python pipeline for cross-dataset benchmarking. 2. An Empirical Generalization Map: Quantifying performance collapse across 5 ML models using MCC, Macro-F1, and FAR. 3. A Feature Alignment Toolkit: Reconciling CICFlowMeter and Argus/Zeek schemas with CORAL. 4. A Classifier Robustness Ranking: Identifying which architectures (tree ensembles vs SVM) best withstand covariate shift. 5. An Interactive Streamlit Dashboard: A production-ready web application for real-time cross-dataset simulation, dynamic confusion matrices, and ROC exploration."

Transition Cue: [Action: Click to Slide 12]

SLIDE 12: 8. References & Closing

Speaker 4: Md Sabique Huda

Visual & Focus: APA citations (Ahmim, Bertoli, Cantone, Sarhan, Verkerken) + Final Thank You.

Spoken Script: "Our project is grounded in leading peer-reviewed literature from IEEE TNSM, IEEE CSR, Software Quality Journal, and IEEE Access. In conclusion, our research takes machine learning intrusion detection out of the closed laboratory world and provides engineered, practical solutions for real-world network deployment. On behalf of Pynskhemlang, Tehjib, Bamelaai, and myself, we express our gratitude to guide Mr. Abdul Kayum Ali sir, supervisor Mr. Debarbat Bharali sir, and the faculty members. We are now open for your questions. Thank you!"

Transition Cue: [Conclusion • Open for Panel Q&A;]

3. Examiner Defense & Key Q&A; Cheatsheet

Essential answers to anticipated questions during the presentation defense:

Q1: In Slide 5, you cite Verkerken et al. (2022) who already tested CICIDS2017 ↔ UNSW-NB15. Isn't your project already done?

Answer: 'Sir/Madam, Verkerken et al. (2022) conducted an observational diagnosis. They documented that accuracy drops 28–42% and concluded that feature harmonization is mandatory, but did NOT build an active solution. As highlighted in our **Slide 7 (Research Gap I: Diagnosis vs. Remediation)**, our project moves from diagnosis to active engineering: we actively construct the semantic ontology, implement unsupervised CORAL domain alignment, evaluate the Generalization Recovery Ratio (GRR), and deploy an interactive Streamlit SOC dashboard. Verkerken diagnosed the disease; our project builds the cure.'

Q2: In Slide 4, why did you choose CICIDS2017 and UNSW-NB15 specifically?

Answer: 'Because they represent the two most widely adopted but structurally incompatible benchmark families globally. CICIDS was captured on a 1 Gbps network using CICFlowMeter (80+ features), whereas UNSW was captured on a 10 Gbps testbed using Argus/Zeek (49 features). Testing transfer between them represents the most authentic simulation of deploying a model into an unseen enterprise network.'

Q3: In Slide 10, how does CORAL alignment work without target labels?

Answer: 'CORAL is unsupervised. It computes the covariance matrices of source and target network flows, whitens the source features, and recolors them with the target covariance. Because it only requires traffic distributions and zero labels, it is completely practical for operational networks where incoming target traffic is unlabeled.'

Q4: Why do you evaluate Matthews Correlation Coefficient (MCC) instead of just Accuracy?

Answer: 'Cybersecurity traffic has extreme class imbalance — over 80% to 85% of flows are benign. A naive model predicting "Benign" for everything achieves 85% accuracy while missing 100% of attacks! MCC considers all four quadrants of the confusion matrix (TP, TN, FP, FN) and gives a balanced score between -1 and +1, accurately reflecting true detection capability.'

Q5: What if domain alignment doesn't recover 100% of accuracy?

Answer: 'A 100% recovery across completely different physical networks is theoretically impossible due to concept drift. What our project proves is how much of the deficit can be recovered through feature harmonization and covariance alignment. Even lifting cross-dataset accuracy from 45% to 75%+ represents a monumental operational victory for production NIDS.'